

Lab 4 — VibeCoding

Maze Runner — Detailed Work & Completion Report

Repository: https://github.com/rohancodexp/12_maze_runner

Project: C:\12_maze_runner

Final environment: Python 3.12.10, pygame 2.5.2, Windows PowerShell virtual environment

1. Purpose

This is a supplementary technical record of the completed individual Lab-4 VibeCoding Maze Runner work. It documents the implementation, validation, Git history, recordings, and remaining submission actions. If the assignment separately requires the official ChatGPT conversation exported as PDF, this report does not replace that chat-history deliverable.

2. Starting Point

The starter Pygame Maze Runner generated a maze, created a player and exit, and ran a basic game loop. The important initial defect was that player collision checked outer boundaries but did not correctly enforce internal maze wall flags, allowing movement through internal walls.

3. Environment

The project originally required pygame==2.5.2. Python 3.14 caused a pygame installation/build problem, so Python 3.12.10 was used successfully. The completed game runs with pygame 2.5.2.

4. Task 1 — Correct Maze Wall Collision

The player collision logic was changed to respect the existing representation `walls[row][column] = [N, S, E, W]`. The player's rectangle is mapped to touched cells and north/south and west/east wall flags are checked while preserving boundary checks. Validation included `py_compile`, running the game, `git diff --check`, and manual wall-collision testing. Commit: 7bad6f5 fix: correct maze wall collision.

5. Task 2 — BFS Shortest-Path Hint

A BFS solver using `collections.deque` was added to `game/maze.py`. It operates on maze cell coordinates, respects the N/S/E/W wall representation, uses a parent dictionary to reconstruct the shortest path, and returns a start-to-goal cell path. H toggles the hint. The path is calculated from the player's current cell to the exit and is recomputed after movement. Reset clears the hint state. Commit: 598ec48 feat: add BFS shortest path hint.

6. Task 3 — Fog of War

Fog of War was added around the player. A transparent Pygame surface is used, with visibility limited to roughly three maze cells around the player. The intended rendering order is maze, path, player, then fog so distant maze regions and the exit can be hidden. Commit: 4d67f0d feat: add fog of war.

7. Task 4 — Timer and JSON Leaderboard

The timer starts when a run begins and stops at the exit. Leaderboard logic was separated into `game/leaderboard.py`. It safely handles missing/corrupt JSON, keeps the best five times in ascending order, saves on a completed run, and displays results on the win screen. Runtime leaderboard data is intentionally ignored by Git. Commit: 76cb749 feat: add timer leaderboard with JSON persistence.

8. Task 5 — Difficulty Tiers

The game was expanded to MENU, PLAYING, and WON states. Easy is 10x8, Medium is 15x13, and Hard is 20x18. Difficulty buttons support hover/click selection; the window is resized for the selected maze. R regenerates the current difficulty and Esc returns to the menu. BFS, fog, collision, timer, and leaderboard behavior were integrated into the difficulty-based flow. Commit: 0d8c159 feat: add difficulty tiers.

9. Demonstration Videos

A BEFORE recording was made using the starter behavior and an AFTER recording was made after all features were implemented. Both were stored under video/ as video/before.mp4 and video/after.mp4. Commit: 5931124 docs: add Lab-4 before and after videos.

10. Repository and Git

The project was maintained with separate commits for the major tasks and pushed to the GitHub repository. A runtime ignore rule was corrected so leaderboard.json and the virtual environment are ignored. The final verification showed the branch up to date with origin/main and the working tree clean.

11. Final Submission Status

Implementation and demonstration work are complete. Remaining work is documentation/submission packaging: verify the GitHub repository and video links, export/share the official VibeCoding ChatGPT conversation as PDF if required, and assemble the final submission information.

12. Difficulty Configuration

Difficulty	Columns	Rows
Easy	10	8
Medium	15	13
Hard	20	18

13. Final Commit History

```
664210b chore: fix runtime data ignore rules
13c3d8c chore: ignore runtime leaderboard data
5931124 docs: add lab-4 before and after videos
0d8c150 feat: add difficulty tiers
76ch740 feat: add timer leaderboard with JSON persistence
4d67f0d feat: add fog of war
598ec48 feat: add BFS shortest path hint
7bad6f5 fix: correct maze wall collision
f059afd Initial Maze Runner starter
```

14. Final Checklist

- All five implementation tasks completed.
- Separate task commits created.
- BEFORE and AFTER recordings completed.
- Videos committed under video/.
- GitHub main branch pushed and synchronized.
- Runtime leaderboard and virtual environment excluded through .gitignore.
- Final working tree clean.
- Official ChatGPT/VibeCoding conversation PDF remains the main chat-history documentation item if required by the handout.
- Final repository, video, and chat-history links should be collected for submission.

Conclusion: Lab 4 implementation and demonstration are complete; only final documentation and submission packaging remain.